

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

I C. Koolakandamthy (192021003) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

C. Koolakandamthy

20/20
20/20

Network throughput and packet loss

* total data received = 8000 bits

* time = 4 seconds

* packet sent = 200

* packet received = 180

* Efficient if throughput > 1500 bps

Pseudo code :

Algorithm network performance:

BEGIN

total data received = 8000

time = 4

packet sent = 200

packet received = 180

IF time = 0 THEN

Display "Error : Time

cannot be zero "

Wait for stop before start

END IF

If throughput > 1500 AND
Packet loss

percentage < 10 THEN

Network Status,
Efficient

ELSE
Network Status,
Inefficient

END IF

Display "Network

Classification :
1 Network Status

END

Pseudocode :

Algorithm

Monitor

Bandwidth

DECLARE

bandwidth wage[]

DECLARE link names[]

number of links = 5

link names [1] = "Link 1"

link names [2] = "Link 2"

link names [3] = "Link 3"

link names [4] = "Link 4"

link names [5] = "Link 5"

while true do

DISPLAY

link names [i] . 1) operating

normally . 1)

DISPLAY

Calculation:-

throughput:

throughput = total data received / time

$$= 8000 / 4$$

$$= 2000 \text{ bps}$$

packet loss percentage:

$$\text{packet loss} = (20 / 2000) = 1\%$$
$$= 1\%$$

Since throughput is 2000 bps which is greater than 1500 bps. But packet loss is 1% and network classification is inefficient.

2)

Bandwidth

utilization

monitoring

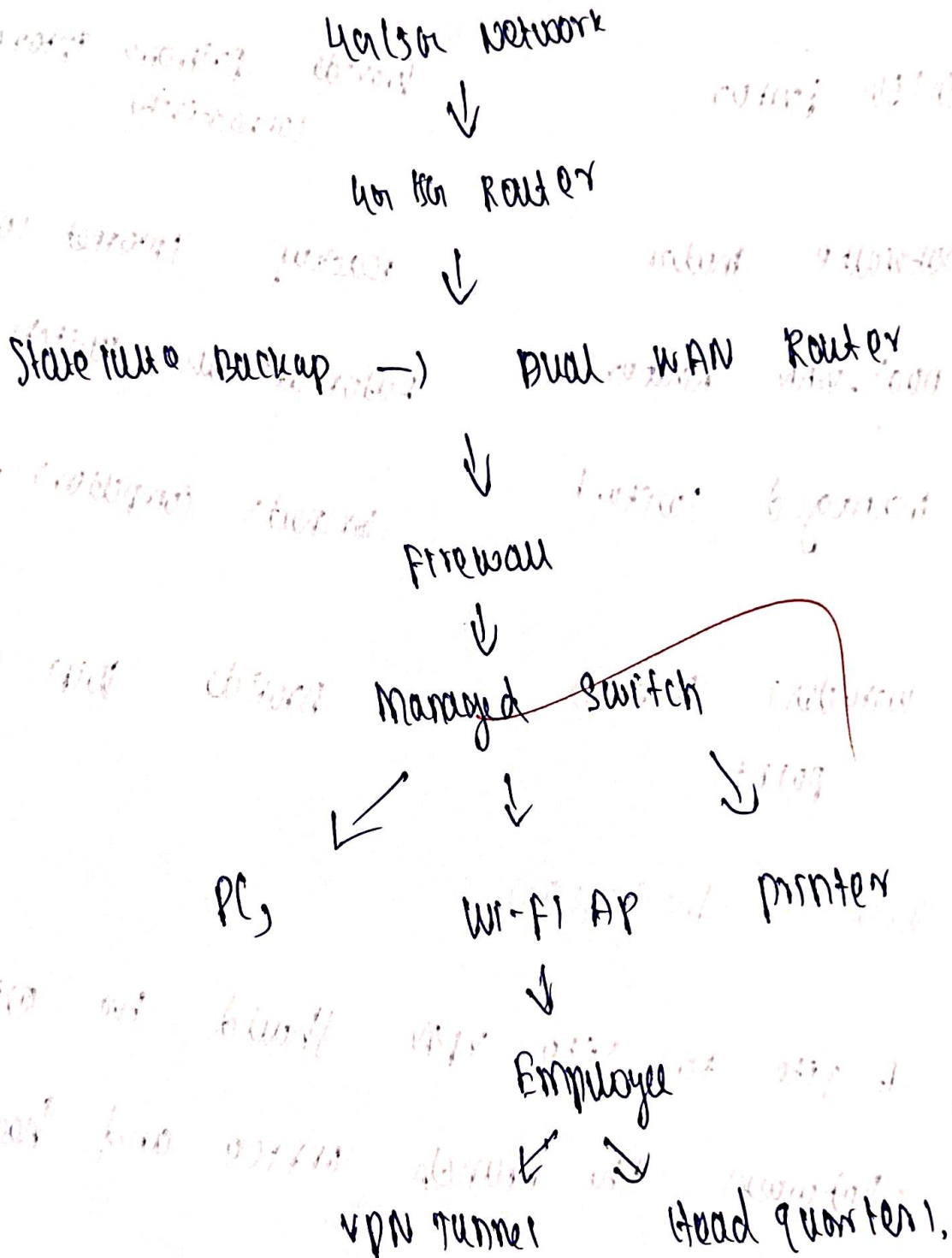
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• Allow only required services

E : Network Architecture



c) Network devices

Device

Purpose

WAN Router

Provide primary Internet connectivity

Subrouter / modem

Backup Internet connection

Multi-WAN Router

Automatically switch connection

Managed switch

Connects computers, printers

Wireless Access points

Provide WiFi connectivity

d) Security Mechanisms

A site to site VPN should be established between the branch office and headquarters

Firewall

Block unauthorized traffic